

Nifty 100 Financial Intelligence Platform

!Platform Banner

Production-Oriented Financial Data Engineering & KPI Analytics Platform for the Indian Equity Market

An end-to-end, local-first analytics pipeline that automates raw financial filing ingestion, enforces 16 strict data validation rules, populates an indexed relational database, and compiles visual equity research insights.

System Architecture • ■■ Database Design • ■ FastAPI Details • ■ Design Decisions • ■■ Dashboard Preview • ■ Test Suite Results

!Python 3.11 / 3.12](https://www.python.org/)

![SQLite 3](https://sqlite.org/)

![FastAPI](https://fastapi.tiangolo.com/)

![Streamlit](https://streamlit.io/)

![Plotly](https://plotly.com/)

![pytest](https://docs.pytest.org/)

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![N100 CI Pipeline](https://github.com/krishnavasani07/n100-financial-intelligence-platform/actions/workflows/ci.yml)

Repository Statistics

■ Companies	■ Financial KPIs	Dashboard Screens	■ API Endpoints	■■■ DQ Ingestion Rules	Automated Tests	■ PDF Reports
92	50+	8 Pages	24 REST Routes	16 Validation Rules	211 Tests (100% Pass)	92 Tearsheets + 11 Sectors

Day 45 Final Technical Release & Sign-Off

The platform has successfully completed the **Day 45 final technical audit** and is signed off as production-ready.

- **Acceptance Gates Audit:** 20/20 Gates PASS (Detailed in `acceptance_checklist.pdf`, with a documented analytic caveat for KMeans clustering silhouette score **0.32**).
- **Deliverable Archive:** 23/23 Deliverables verified and compiled under `output/final_deliverables/`.
- **Automated QA Suite:** 211/211 tests passing with 100% success rate.

- **Submission Materials:**
- **Final Project Report:** `final_project_report.pdf` (Comprehensive 13-section technical report).
- **Presentation Slides:** `final_presentation.pdf` (10-slide landscape presentation deck).
- **Demo Video script:** `demo_video_guidelines.md` (Walkthrough narration structure).
- **Submission Audit Checklist:** `submission_audit.md` (Compliance audit report).

What Is This?

This project builds an end-to-end financial intelligence pipeline for the Nifty 100 universe. It ingests structured financial data from raw filing formats, normalizes and validates it through 16 data-quality rules, stores it in a locally indexed SQLite database, computes 50+ financial KPIs, exposes analytics through a FastAPI REST backend, and presents the results through an interactive Streamlit research dashboard.

Project at a Glance

Layer	Technology Stack	Primary Responsibility
Ingestion	Python / Pandas / OpenPyXL	Auto-detect sheet offsets, standardize ticker casings, and load raw filing data.
Validation	Custom Data Quality Engine	Evaluate 16 data-quality validation rules prior to ingestion.
Storage	SQLite 3 (WAL Mode)	Relational persistence with foreign keys and multi-index lookups.
Analytics	Pandas / NumPy / Scikit-Learn	Compute 50+ financial KPIs, CAGR growth metrics, and execute KMeans clustering.
API	FastAPI / Uvicorn	Expose versioned REST endpoints with autogenerated interactive Swagger documentation.
Dashboard	Streamlit / Plotly	Render interactive charts, search overlays, multi-slider screeners, and watchlists.
Reports	ReportLab / Matplotlib	Compile batch PDF tearsheets and multi-page sector benchmark reports.
Testing	Pytest / Pytest-HTML	Layer-based unit, API, and integration test coverage verifying code correctness.

Problem & Motivation

The Problem

Raw corporate financial filings in the Indian equity market are highly fragmented and dirty:

1. **Format Drift:** Excel structures, column names, and row offsets fluctuate between reporting quarters and years.

2. **Ticker Mismatches:** NSE and BSE symbols are represented with varying suffixes, spaces, or casings (e.g., **TCS.NS**, **Tcs**, **TCS**).
3. **Arithmetic Anomalies:** Balance sheets occasionally fail to balance ($\text{\$Assets} \neq \text{\$Liabilities}$) due to missing rows, and reported margins frequently mismatch computed margins.
4. **NoSQL Limitations:** Ingesting raw dumps without strict schema verification leads to downstream calculation failures.

The Solution

Instead of charging tens of thousands of dollars for institutional terminals or relying on closed-source retail screeners, this platform demonstrates how standard software engineering practices—such as transactional schema loading, mathematical growth rate guards, decoupled validation, and automated unit testing—can build a robust, transparent, and local-first research engine.

Key Features

- **Multi-Page Streamlit Dashboard:** 8 comprehensive, independent financial analytics pages routed through a native sidebar navigation menu.
- **Interactive Plotly Visualizations:** High-fidelity, hover-responsive charts including sector bubble plots, 10-year YoY growth trends, and capital allocation treemaps.
- **Cached Database Layer:** Connection caching (**@st.cache_data**) and SQLite Write-Ahead Logging (WAL) mode enabling sub-millisecond query responses and page load speeds.
- **Institutional Valuation Engine:** Automated valuation pipeline computing FCF Yields and Sector Median P/E multiples to flag stocks under **Discount** or **Caution** categories.
- **Responsive Multi-Slider Filtering:** Instant screener updates using custom presets that automatically populate 10 sliding filters via Streamlit session states.
- **Automated Excel & CSV Exports:** Reusable download widgets for dynamic screeners, alongside structured openpyxl reports compiling relative valuations.
- **Custom Financial Math Engines:** Computes profitability (DuPont, ROE/ROCE), CAGR growth curves, and cash flow structures with built-in mathematical edge-case guards.
- **Batch PDF Research Report Pipeline:** Programmatic ReportLab compiler generating styled 2-page company tearsheets and multi-page sector summary reports. Implements data eligibility controls and layout QA validations.
- **FastAPI REST API Layer:** Clean versioned endpoints providing programmatic database access for external scripts, tools, and clients.

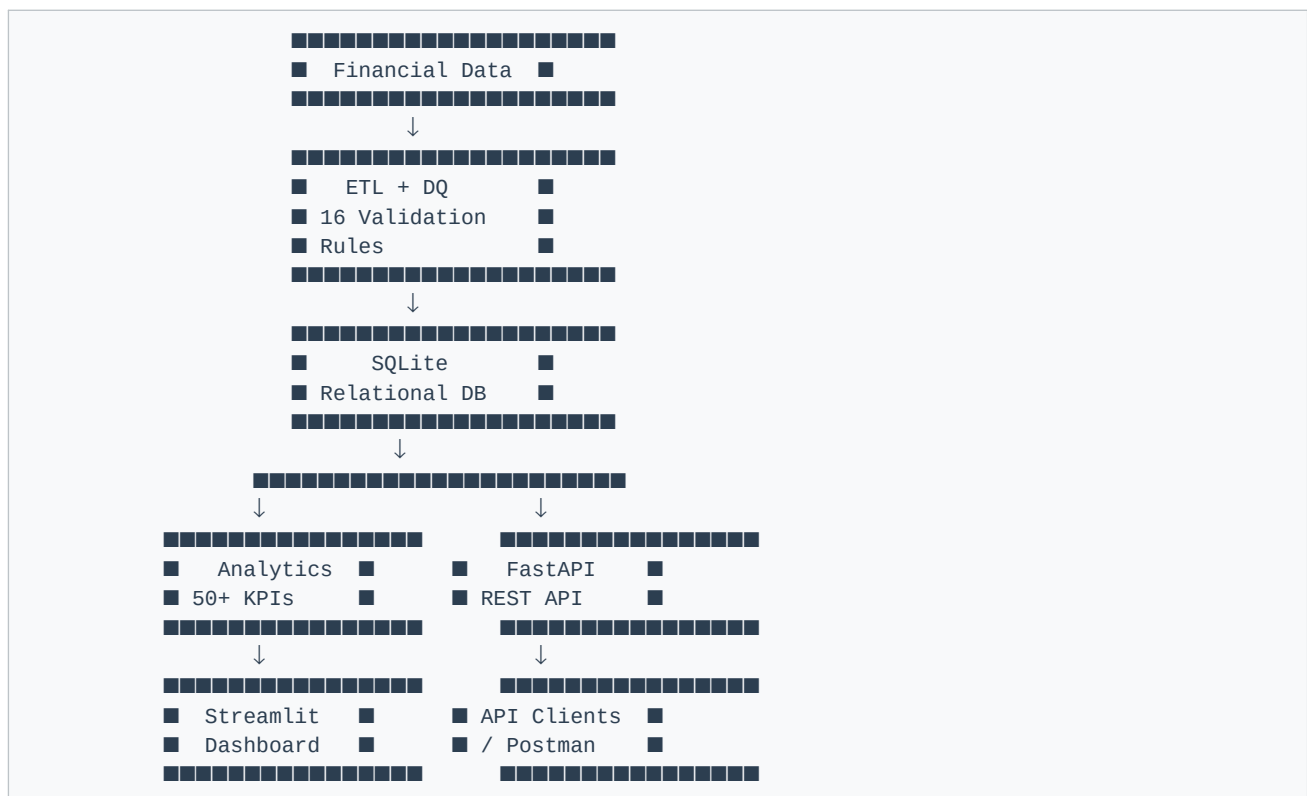
Engineering Highlights

- **Transactional SQLite Ingestion:** Ingests raw data using single-session SQLite database transactions, ensuring automatic rollbacks on ingestion failure to prevent schema corruption.
- **Write-Ahead Logging (WAL):** Configures SQLite in WAL mode to support parallel read actions and reduce database access latencies.

- **Deterministic KPI Calculations:** Uses strict mathematical formulas with built-in growth division-by-zero guards to prevent NaN outputs.
- **Reproducible Clustering:** Performs KMeans peer clustering using a fixed random state for predictable grouping.
- **Data-Quality Validation:** Decouples 16 schema and integrity rules from direct SQL load, creating detailed error CSV logs on validation failure.

■■ Architecture & Data Flow

The platform separates ingestion, quality assurance, storage, mathematical computation, and client display layers:



■ End-to-End Data Flow

```

flowchart TD
    subgraph Ingestion_Layer [1. Ingestion Layer]
        A[Excel filings / raw data] --> B[Inbound Parser]
        B --> C[ExcelLoader Engine]
        C --> D[Date & Ticker Standardizer]
        D --> E[String Normalization]
    end

    subgraph Data_Quality_Suite [2. Validation Engine]
        F[16 DQ Rules Evaluator] --> G[5 Critical Blockers]
        G --> H[Rejection Logs & CSVs]
        H --> I[11 Non-Blockers]
        I --> J[Anomalies Log & Ingestion]
    end

    subgraph Relational_Database [3. Relational DB]
        K[Transactional Load] --> L[(SQLite Relational DB)]
        L --> M[PRAGMA foreign_keys = ON]
        M --> N[Auto-Backup Handler]
        N --> O[(db/backups/)]
    end

    subgraph Analytical_Engines [4. Calculation Engines]
        P[Profitability & Solvency Engine] --> Q[CAGR Growth Engine]
        Q --> R[Cash Flow & Capital Allocation Engine]
        R --> S[Ratios Table]
        S --> T[financial_ratios]
        T --> U[CAGR Output]
        U --> V[growth_summary.csv]
        V --> W[Allocation Matrix]
        W --> X[capital_allocation.csv]
    end

    subgraph API_UI_Clients [5. API & UI Clients]
        Y[Streamlit UI Dashboard] --> Z[FastAPI REST Engine]
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        YD --> YE[0]
        YE --> YF[0]
        YF --> YG[0]
        YG --> YH[0]
        YH --> YI[0]
        YI --> YJ[0]
        YJ --> YK[0]
        YK --> YL[0]
        YL --> YM[0]
        YM --> YN[0]
        YN --> YO[0]
        YO --> YP[0]
        YP --> YQ[0]
        YQ --> YR[0]
        YR --> YS[0]
        YS --> YT[0]
        YT --> YU[0]
        YU --> YV[0]
        YV --> YW[0]
        YW --> YX[0]
        YX --> YY[0]
        YY --> YZ[0]
        YZ --> ZA[0]
        ZA --> ZB[0]
        ZB --> ZC[0]
        ZC --> ZD[0]
        ZD --> ZE[0]
        ZE --> ZF[0]
        ZF --> ZG[0]
        ZG --> ZH[0]
        ZH --> ZI[0]
        ZI --> ZJ[0]
        ZJ --> ZK[0]
        ZK --> ZL[0]
        ZL --> ZM[0]
        ZM --> ZN[0]
        ZN --> ZO[0]
        ZO --> ZP[0]
        ZP --> ZQ[0]
        ZQ --> ZR[0]
        ZR --> ZS[0]
        ZS --> ZT[0]
        ZT --> ZU[0]
        ZU --> ZV[0]
        ZV --> ZW[0]
        ZW --> ZX[0]
        ZX --> ZY[0]
        ZY --> ZZ[0]
    end

    style A fill:#2d3748,stroke:#4a5568,stroke-width:1px,color:#fff
    style G fill:#1a365d,stroke:#2b6cb0,stroke-width:2px,color:#fff
    style O fill:#2c5282,stroke:#4299e1,stroke-width:2px,color:#fff
    style P fill:#005c53,stroke:#009688,stroke-width:2px,color:#fff
    style D fill:#744210,stroke:#d69e2e,stroke-width:1px,color:#fff

```

Component Responsibilities:

1. **src/etl/loader.py**: Auto-detects table offsets, cleans headers, and standardizes ticker formats.
2. **src/validation/validator.py**: Executes the 16 data quality rules, preventing database pollution.
3. **src/database/loader.py**: Loads validated datasets in single transactions with automated database rollbacks if errors occur.
4. **src/analytics/**: Relies on specific calculations (ratios, CAGR, cash flow patterns, and valuation metrics) to populate relational tables.

5. **src/api/**: Serves versioned JSON responses to clients, enabling external script integrations.
6. **src/reports/**: Compiles PDF tearsheets and sector summary booklets with visual layout budgets.
7. **app.py**: Serving the user interface utilizing Streamlit and Plotly visualizations.

Analytics & KPI Engine

The analytics engine processes raw corporate filings to generate normalized financial metrics:

- **DuPont Engine**: Breaks down Return on Equity (ROE) into Net Profit Margin, Asset Turnover, and Equity Multiplier to identify the operational driver behind company returns.
- **Growth Suppressor**: Turnaround periods (transitioning from deep losses to minor profits) are flagged, and CAGR calculations are suppressed when mathematically invalid, preventing erroneous positive growth spikes.

■ KPI Methodology

We calculate and verify key performance indicators (KPIs) following standard financial math guidelines:

- **ROE (Return on Equity)**: $\frac{\text{Net Income}}{\text{Average Shareholders Equity}}$
- **ROCE (Return on Capital Employed)**: $\frac{\text{EBIT}}{(\text{Total Assets} - \text{Current Liabilities})}$
- **Operating Profit Margin (OPM)**: $\frac{\text{Operating Profit}}{\text{Revenue}}$
- **Debt-to-Equity (D/E)**: $\frac{\text{Total Debt}}{\text{Total Shareholders Equity}}$ (suppressed for the *Financials* sector)
- **Interest Coverage Ratio**: $\frac{\text{EBIT}}{\text{Interest Expense}}$
- **Revenue / PAT / FCF CAGR**: Compound Annual Growth Rate measured over 3, 5, and 10-year windows.
- **Valuation Categories**: Stocks classified into *Discount*, *Fair*, or *Caution* based on relative P/E vs. Sector Medians and historical FCF Yields.

For detailed formulas, math definitions, and edge-case guards, refer to:

- **financial_formulas.md**

■■ Data Quality Validation

Prior to database insertion, all datasets must pass 16 checks grouped into five layers:

Category	Checks	Description / Action
Schema Integrity	5	Verifies column mappings, data types, standard headers, and table structures. <i>Critical Blocker.</i>
Financial Consistency	4	Confirms Balance Sheet balances (\$Assets = Liabilities) and checks for logical relationships. <i>Critical Blocker.</i>

Missing Data	3	Traces null values, gap detection for missing years, and flags incomplete data. <i>Non-blocker.</i>
Referential Integrity	2	Maps tickers to master metadata lists and checks sector allocations. <i>Critical Blocker.</i>
Business Rules	2	Validates that reporting periods are logical and prevents division by zero in ratios. <i>Non-blocker.</i>
Total Rules	16	<i>5 rules act as Critical Blockers (rejects filing), while 11 generate anomaly logs.</i>

For the rule specifications, refer to **etl_pipeline.md**.

■ API

FastAPI exposes the platform's analytical data through a versioned REST API.

- Base URL: **<http://localhost:8000/api/v1>**
- Interactive Documentation (Swagger UI): **<http://localhost:8000/docs>**
- Alternative Documentation (ReDoc): **<http://localhost:8000/redoc>**
- OpenAPI Schema: **openapi.json**
- Postman Collection: **n100_api.postman_collection.json**

Endpoints Table

Method	Endpoint	Description
GET	/health	API and database connection health check, plus row counts.
GET	/companies	List all companies in the Nifty 100 universe.
GET	/companies/{ticker}	Comprehensive company profile metadata and details.
GET	/companies/{ticker}/pl	Historical Profit & Loss table.
GET	/companies/{ticker}/bs	Historical Balance Sheet table.
GET	/companies/{ticker}/cashflow	Historical Cash Flow table.
GET	/companies/{ticker}/ratios	Historical KPI and financial ratio tables.
GET	/companies/{ticker}/tearsheet	Programmatic link to download company tearsheet PDF.
GET	/screener	Multi-factor stock screening with custom thresholds.
GET	/sectors	Summary statistics (OPM, ROE, P/E) across 11 sectors.
GET	/sectors/{sector}/companies	List all constituents in a specific sector.
GET	/peers/{group_name}	Peer analytics and sector medians comparisons.
GET	/companies/{ticker}/peers/compare	Dynamic relative performance comparisons.
GET	/valuation	Sector-relative valuation summary.
GET	/market-cap/{ticker}	Valuation categorization (Discount, Fair, Premium).
GET	/portfolio/stats	Platform-wide percentile metrics for financial benchmarks.
GET	/companies/{ticker}/documents	Retrieve paths to annual filings and PDF reports.

API Request/Response Example

Request:

```
curl -X "GET" "http://localhost:8000/api/v1/screener?min_roe=15" -H "accept: application/json"
```

Response:

```
{
  "count": 54,
  "results": [
    {
      "ticker": "TCS",
      "company_name": "Tata Consultancy Services Limited",
      "sector": "Information Technology",
      "roe_pct": 48.2,
      "debt_to_equity": 0.02,
      "interest_coverage": 89.4
    }
  ]
}
```

Dashboard Preview

The frontend is a local multi-page Streamlit web app displaying live financial distribution matrices and valuation charts:

!Streamlit Executive Dashboard

The application is broken down into **8 comprehensive, independent pages** allowing granular research:

■ 1. Executive Home

!Executive Home

Provides high-level market summaries, sector distribution treemaps, and quality-score rankings across the Nifty 100 universe.

■ 2. Company Profile

!Company Profile

Displays a deep-dive financial profile for a selected company, detailing profitability margins, DuPont analysis, leverage metrics, and watchlists.

■ 3. Investment Screener

!Investment Screener

Includes 10 interactive sliders and 6 predefined investment strategy presets (e.g., Value Pick, Dividend Champion) with real-time filtration and CSV export.

■ 4. Peer Comparison

!Peer Comparison

Enables head-to-head financial ratio comparisons and sector peer benchmarking using interactive, normalized radar charts.

■ 5. Trend Analysis

!Trend Analysis

Plots 10-year historical trajectory charts for individual corporate financials (sales, profits, assets) alongside Year-over-Year (YoY) percentage changes.

■ 6. Sector Analytics

!Sector Analytics

Visualizes industry structures through interactive bubble plots comparing revenue and ROE, with bubble sizes scaled by NSE index weights.

■ 7. Capital Allocation

!Capital Allocation

Classifies companies into strategic categories (e.g., Debt-Free, Capital Efficient, Dividend Leaders) and presents them in a nested hierarchical tree map.

■ 8. Reports Browser

!Reports Browser

Allows analysts to search, view, and directly download PDF copies of corporate annual reports and financial filings stored locally.

■ Automated Reports

The reporting engine generates publication-quality PDF assets compiled under the **reports/** folder:

1. **Company Tearsheets:** Clean 2-page documents summarizing historical ratios, DuPont operational breakups, cash flow waterfalls, and capital allocation badges.
2. **Sector Summary Booklets:** Performance digests compiling median trends, sector performance indicators, and peer ranks.

Both types of reports implement layout QA constraints (strict size limits and page budgets) and require a minimum of 3 years of financial data before compiling.

■ Technology Stack

- **Language:** Python 3.11 / 3.12
- **Data Processing:** Pandas, NumPy, OpenPyXL, PyYAML
- **Relational Storage:** SQLite 3
- **API Layer:** FastAPI, Uvicorn, Pydantic
- **Interactive UI:** Streamlit, Plotly, Pillow
- **Reporting Output:** ReportLab, Matplotlib
- **Testing Engine:** Pytest, Pytest-HTML, HTTPX

■ Repository Structure

```
n100-financial-intelligence-platform/
■■■■ config/                # Screener YAML rules and threshold parameters
■■■■ data/                  # Local raw and processed datasets (git-ignored)
■■■■ db/                    # SQLite DB store, schemas, and backups
■   ■■■■ schema.sql         # Database schema definition
■   ■■■■ backups/           # Rotation backups directory
■■■■ docs/                  # Architecture, system, and formulas specs
■■■■ notebooks/             # Interactive exploration & prototyping notebooks
■■■■ output/                # Log outputs, PDF exports, and ratio CSV logs
■   ■■■■ perf_notes.md      # Day 43 Performance Testing Report
■■■■ README_ASSETS/         # Banner and dashboard preview files
■■■■ scratch/               # Local debug and ad-hoc scripts
■■■■ scripts/               # Performance benchmark and automation scripts
■   ■■■■ benchmark_profile_load.py # Profile latency benchmarking script
■   ■■■■ load_test_screener.py  # Concurrent API load testing script
■■■■ src/                   # Core Application Source Code
■   ■■■■ analytics/         # Ratios, CAGR, cash flow, and valuation engines
■   ■■■■ api/               # FastAPI router logic, models, and middleware
■   ■   ■■■■ routers/       # Modular router files for endpoints
■   ■   ■■■■ database.py     # FastAPI database connection pools
■   ■   ■■■■ middleware.py   # Request and execution logging middleware
■   ■■■■ config/            # Centralized environment settings
■   ■■■■ dashboard/         # Multi-Page Streamlit App and layout modules
■   ■■■■ database/          # Database loaders and queries
■   ■■■■ etl/               # Parsing and normalization scripts
■   ■■■■ peer_analysis/     # Peer ranks and sector averages
■   ■■■■ reports/           # ReportLab templates & batch PDF pipeline
■   ■■■■ screener/          # Preset filtering and ranking algorithms
■   ■■■■ validation/        # Ingestion rule checkers
■■■■ tests/                 # pytest unit, API, and integration suite
■■■■ app.py                 # Streamlit web application entrypoint
■■■■ Dockerfile             # Production container config
■■■■ LICENSE                # MIT License
■■■■ main.py                # ETL Pipeline runner
■■■■ requirements.txt        # Cleaned, minimal dependencies manifest
```

Quick Start

Follow these commands to clone, configure, and start the application locally.

1. Installation

```
# Clone the repository
git clone https://github.com/krishnavasnani07/n100-financial-intelligence-platform.git
cd n100-financial-intelligence-platform

# Initialize virtual environment
python -m venv .env
.venv\Scripts\activate # Windows
# source .env/bin/activate # macOS / Linux

# Install dependencies
pip install -r requirements.txt
```

2. Startup Commands

```
# Set PYTHONPATH to project root
$env:PYTHONPATH="." # Windows (PowerShell)
# export PYTHONPATH="." # macOS / Linux

# Run ETL Pipeline to populate database
python main.py

# Launch FastAPI Backend
uvicorn src.api.main:app --host 127.0.0.1 --port 8000 --reload

# Launch Streamlit Dashboard (Open a new terminal)
streamlit run src/dashboard/app.py --server.port 8501
```

Once running, access the services:

- **Streamlit Dashboard:** <http://localhost:8501>
- **FastAPI Docs (Swagger):** <http://localhost:8000/docs>

■■ Configuration

Copy the sample configuration file template **.env.example** to **.env** in the root folder:

```
ENV=development
DEBUG=True
DB_PATH=db/nifty100.db
LOG_LEVEL=INFO
LOG_FILE=logs/app.log
```

■ Docker

Start the entire stack (FastAPI on port 8000, Streamlit on port 8501) inside Docker containers:

```
# Configure env
cp .env.example .env

# Build and start services
docker compose up -d

# Check running status
docker compose ps
```

Testing

The platform features automated tests validating ingestion parsers, ratio mathematics, API routers, and database consistency.

Running Tests

```
# Execute test suite
pytest tests/ -v
```

Latest Test Run Summary

- **Tests Collected:** 211
- **Passed:** 211
- **Failed:** 0
- **Test HTML Report:** `pytest_report.html`

Performance

The following benchmarks represent actual measurements taken during Day 43 verification runs.

Test	Objective	Target	Actual	Result
Concurrent API Load	10 concurrent screener requests	Batch time < 10.0s	0.323 seconds	PASS
Profile Load (Cold)	Ingest database values per ticker	Load time < 3.0s	~18 ms	PASS
Profile Load (Warm)	Streamlit <code>@st.cache_data</code> in-memory	Load time < 3.0s	~3.5 ms	PASS
Port Execution	Simultaneous ports 8000 & 8501	No conflicts	No conflict	PASS

For details, log statements, and bottlenecks analysis, refer to `perf_notes.md`.

■ Design Decisions & Trade-Offs

1. Upfront Rule-Based Ingestion vs. DB Constraints:

- *Decision:* Implemented a decoupled validation framework (`src/validation/validator.py`) executing rules on raw DataFrames before writing to SQL.
- *Trade-Off:* Increases memory overhead but prevents database pollution, creating detailed anomalies CSV logs instead of dropping records with generic DB insert errors.

2. SQLite WAL Mode & Threading:

- *Decision:* Selected SQLite with Write-Ahead Logging (WAL) and foreign key enforcement over client-server engines like PostgreSQL.
- *Trade-Off:* Limits database clustering capabilities but guarantees zero-configuration, single-file portability with fast query times for local execution.

3. Growth (CAGR) Math Suppressor:

- *Decision*: Suppressed growth rate calculations for turnaround periods, replacing mathematical NaN values with detailed text strings.
- *Trade-Off*: Hides visual trend lines for volatile companies, but prevents false positive CAGR projections for firms transitioning from heavy losses to small gains.

■ Known Limitations

- **Static Ingestion Templates**: The ETL pipeline expects Excel files structured in tabular form matching our standard schemas. A minor change in sheet layouts requires coordinate overrides.
- **Write Concurrency**: SQLite's WAL mode supports parallel read actions, but write locks remain sequential. Not suitable as a multi-user transactional platform.
- **Local Copilot Mocking**: To run local-first, the "AI Copilot" page uses a structured financial template matching the metrics instead of executing online LLM calls.

■ Development Roadmap

- **Sprint 1 — Ingestion Foundation**: Set up raw Excel parsing and string normalization pipelines.
- **Sprint 2 — Ratios Engine**: Implemented DuPont formulas, CAGR calculations, and division-by-zero guards.
- **Sprint 3 — Advanced Analytics**: Built KMeans peer grouping and sector median calculators.
- **Sprint 4 — Dashboard UI**: Designed the Streamlit multi-page interface and Plotly components.
- **Sprint 5 — Reports Engine**: Programmed the ReportLab PDF compiler for tearsheets and booklets.
- **Sprint 6 — API & QA Verification**: Integrated FastAPI routers, added the 211-test suite, and completed Day 43 performance load-testing.

■ Future Work

- **P1 — PostgreSQL Integration**: Add database connection adapters to scale the database for multi-user hosting.
- **P1 — OCR Filing Parser**: Integrate unstructured PDF text parsers to ingest annual reports directly.
- **P2 — Live Price Feeds**: Connect a daily stock market scraper to refresh current valuations dynamically.
- **P2 — Vector Store RAG Copilot**: Build a localized retrieval-augmented generation engine to query footnotes in annual filings.
- **P3 — Cloud Deployment**: Set up automated Terraform configurations for AWS or GCP cloud staging.

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